Kooya PH AI Research

Computational biology models for the Philippines — capability brief and introduction to prospective wet-lab partners

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1. The company

Kooya PH AI Research is an independent group building artificial-intelligence models for biology, focused on problems that matter to the Philippines: agricultural biotechnology, industrial enzymes, and public health. [Add one or two lines: where you are based, how the group is organized, any affiliations.]

Our working principle is honesty over hype. Everything described here is **computational and at the research stage**. None of our designs has been tested in a laboratory yet, and we take care never to present a prediction as an experimental result. This document is deliberately detailed, including our negative findings, because we would rather a partner understand exactly where we are than be oversold.

2. Who we are

[Your name, role, and background — one to three lines.] We operate as a focused team that combines protein and chemistry modeling with modern AI engineering for data pipelines, training, and evaluation. What we do not have, and what we are seeking, is a wet-lab partner who can tell us what is realistic at the bench and, when the time is right, test our strongest candidates.

3. Our models, in detail

For each model we give what it is, its objective, where it is useful, what we have actually observed, and its current limitations.

FLORA — protein function analysis (operational)

What it is. A classifier that reads a protein sequence and assigns a coarse functional class — enzyme, transport, storage, structural, or other — using a large pretrained protein language model as a frozen encoder with a trained classification head on top.

Objective. Give researchers a fast, sequence-only functional read on crop and food proteins so that limited laboratory effort is spent on the most promising candidates.

Use cases. Triaging a batch of uncharacterized plant proteins by likely function; first-pass annotation in crop and food proteomics; a sanity filter that flags when a designed or unknown sequence does not look like what it is supposed to be.

What we have observed. On our internal benchmark the model reaches a macro-F1 of about 0.795 across the five classes, consistent across five training seeds (spread of about 0.003). The test proteins were chosen so that none has a close relative (30% sequence identity or more) in the training set. We found that roughly 93% of the improvement over our earlier version came from a single modeling choice, how the sequence representation is pooled, rather than from any deep advance. We also tried adding a fine-tuning adapter on top; it did not clear the improvement threshold we had pre-registered, so we did not adopt it. We report this openly rather than claim a breakthrough.

Limitations. Analysis only: it does not generate, fold, or predict catalytic activity. It is validated on one benchmark and one task family; its output score ranks the classes but is not a calibrated probability of being correct; its scope is crop and food proteins. We make no state-of-the-art claim.

PH163 — variant-effect scoring (operational)

What it is. A tool that estimates how disruptive a single amino-acid substitution is in an existing protein, using an established zero-shot method (masked-marginal scoring) on a large protein language model — no task-specific training required.

Objective. Help prioritize which mutations, among many, are worth the cost of making and testing, with a particular interest in Philippine rice.

Use cases. Ranking the variants in a mutagenesis panel; scoring natural variation across Philippine rice accessions (a planned, pre-registered study, AGRI-RICE, would compare 163 local accessions against a 3,000-genome background) to focus attention on the changes most likely to matter.

What we have observed. We have confirmed that the service runs end to end and returns scores. We have not yet run a formal benchmark against labeled mutation datasets, so we cannot say how accurate the scores are; we treat them strictly as a prioritization signal, not a verdict.

Limitations. The score is a model estimate, not a calibrated probability or a phenotype prediction, and it makes no claim of biological causation. It handles one substitution at a time and is not yet formally benchmarked.

LIKHA — protein sequence generation (operational, under active specialization)

What it is. A 354-million-parameter generative protein model we trained from scratch on plant and industrial proteins. Given a prompt, it writes new amino-acid sequences one residue at a time. It ships with two fast internal screens: a foldability probe (which estimates, without running a full structure prediction, whether a sequence looks structurally plausible) and an exact-match check against its own training data.

Objective. Produce novel, plausible candidate enzyme sequences for Philippine agri-waste — specifically lignocellulose-degrading enzymes (cellulases, xylanases, beta-glucosidases) that break down rice straw — as starting points for enzyme discovery and engineering.

Use cases. Proposing candidate enzyme sequences for rice-straw valorization; seeding diverse sequence libraries for downstream screening; providing engineering starting points when a scaffold or family is known.

What we have observed (this is where we have the most experimental detail). - *Sampling matters a great deal.* By tuning the sampling temperature and nucleus cutoff, the fraction of generated sequences that a structure predictor (ESMFold) folds with high confidence (pLDDT of at least 70) rose from roughly 8% to about 37% in our test. This was measured on a small sample, so we treat the exact figure as approximate. Separately, our fast foldability probe separates real proteins from scrambled ones very reliably, and remains useful (though less sharp) when applied to the model's own generated sequences. - *The stage-1 model memorizes part of its training data.* When we prompted it with training proteins, its greedy continuations reproduced the training sequences in most cases, so we screen explicitly for exact reproduction and report novelty rather than assume it. - *Family specialization improves the model on held-out enzymes.* After adapting the model on 555 reviewed cellulase, xylanase, and beta-glucosidase sequences (the validation set was held out by sequence-similarity clustering to prevent leakage), perplexity on the held-out enzymes fell by roughly 10%, with a confidence interval that excludes zero, while general-protein modeling stayed essentially unchanged. The improvement also holds on the subset of validation sequences least similar to the training set. - *A methodology error we caught.* Our first adaptation runs silently skipped long sequences (most of these enzymes are longer than the 400-residue limit we had set), which made them unreliable. We corrected the limit and re-ran; every number in this document comes from the corrected runs, and we have set aside the earlier conclusions. - *Our family prompts behave as a general "enzyme switch," not yet a family switch.* In a controlled test, 20 of 480 prompted sequences aligned to known enzymes in our reference set, versus 0 of 360 unprompted ones: a statistically clear effect that the prompt pushes generation toward enzyme-like output. But only 5 of those 20 aligned to the *requested* family, no better than chance, and the loss-level advantage of the correct prompt over a wrong one is real but very small (about 0.3% in perplexity). We therefore cannot claim on-demand control of a specific enzyme family. - *Some enzyme-like outputs are near-copies of training data.* Six of those 20 aligned sequences were 77-85% identical to training enzymes, mostly beta-glucosidases, regardless of which prompt was used. Alignment to a known enzyme is therefore not, by itself, evidence of novelty, and we assess the two separately. - *Raw output quality is still limited.* Only about one in five generated sequences terminates naturally, and roughly half of raw generations are repetitive or low-complexity; we now filter these out before ranking. Of 840 distinct generated sequences, 820 had no detectable alignment to our reference enzymes, which is expected for a sparse reference set but means we cannot yet say what most of them are.

Limitations. This is an early (stage-1) model. Its prompts do not yet reliably produce a specific target family; part of its raw output is low quality and discarded; and every surviving sequence is a computational candidate, never a demonstrated enzyme. We are expanding the training data to address the family-control and quality gaps, which is exactly why we are not sending candidates for testing today.

HYGEIA — diagnostic protein-binder design (in development, planned)

What it is (planned). A design system that would create new protein binders — proteins shaped to stick to a chosen disease-target protein — for use as the detection reagents inside cheap rapid diagnostic tests. It would build on published, openly licensed structure-based design tools rather than training a structure predictor from scratch.

Objective. Design locally made detection reagents for diseases that burden the Philippines, beginning with dengue and tuberculosis, to reduce dependence on imported test components. This is detection, not treatment.

Use cases. Capture and detector proteins for dengue and TB rapid tests; in general, a binder against a well-characterized antigen that can be validated at the bench with a binding assay in months rather than the years a therapeutic would take.

What we have done so far. A desk review of the two leading open design tools we are considering, covering their licenses, how their model weights are distributed and in what file format, their software requirements, and the authors' stated responsible-use measures. We have not downloaded or run either tool, and no design work has begun. Choosing which pathogen protein to target has not been done and is held for governance review.

Limitations. Not started. The tools must still pass a hands-on security audit before we run them, and because it targets pathogen proteins — even for benign detection — it proceeds only after institutional biosafety clearance and our own governance review. Predicted binding is a hypothesis until a laboratory binding assay confirms it.

Molek — small-molecule generation (in development, planned)

What it is (planned). A generative model for drug-like small molecules, about 30 million parameters, built on discrete graph-diffusion and flow-matching methods and trained on our own weights from scratch. We do not claim to have invented the architecture: each stage first reproduces a published open recipe as a control, so that a reference always exists to check that our implementation is correct, and only then trains ours. Molek is the "propose candidate molecules" stage of a larger pipeline: target protein, then pocket and binding-affinity estimate (an open-source scorer, Boltz-2, which we use but do not train), then our generator, then absorption and toxicity predictors (open-source scorers), then a safety screen.

Objective. Propose candidate small molecules against Philippine-priority disease targets, starting with tuberculosis and dengue, as the front end of a longer therapeutic-discovery effort.

Use cases. Generating candidate molecules around a known target or scaffold; hit-to-lead expansion; seeding chemical libraries for virtual screening.

The staged plan. - *Stage 0, reproduce.* Run the published open recipes unchanged on the standard MOSES benchmark and confirm we match their reported results. - *Stage 1, Molek v2 base.* Train our roughly 30-million-parameter 2D generator on MOSES plus license-clean additions, deduplicated against the benchmark test sets, and evaluate with pre-registered metrics: validity, uniqueness, novelty, distribution similarity (FCD), and scaffold diversity. - *Stage 2, pocket-conditioned generation.* The actual structure-based drug-design generator, a pocket-conditioned 3D diffusion model trained from public protein-ligand complex data, built only after Stage 1 works. Evaluation: physical-validity checks, rescoring of binding affinity with Boltz-2, and docking, compared with published reference checkpoints. The reference recipe uses four GPUs for many epochs, so we will measure the cost on one part first; it may not fit our weekly compute budget in one piece. - *Stage 3, domain targets.* Choose health targets with our research reviewers and add absorption and toxicity predictors as property scorers.

Candidate targets (none selected yet). From the literature and the public structure database: for tuberculosis, InhA, DprE1, DNA gyrase (GyrA/GyrB), ATP synthase, QcrB, and MmpL3; for dengue, the NS2B-NS3 protease and the NS5 polymerase are the realistic pocket-based candidates. The dengue NS3-NS4B interaction, the mechanism of a published dengue antiviral candidate (mosnodenvir), is a protein-protein interface and fits a pocket-conditioned generator poorly. Target proteins would be used only as inference-time inputs; the generator itself would be trained on general protein-ligand complexes. The structure counts we pulled from the public database used loose matching, so they are upper bounds that still need manual curation.

What we have done so far. As of our latest written blueprint (5 November 2026), planning only: we read the open recipes' configurations and licenses and mapped candidate targets, but nothing has been trained, no datasets downloaded, and no compute spent. [Stan: confirm this is still accurate before sending; update it if training has started.]

Planned safeguards. Known chemical-warfare agents and close analogues would be removed from training data and every output screened against the same list; toxicity predictors would never be used as a reward to be maximized; and the model would stay internal behind a gated, audited access tier. Every adopted tool must pass a security and license audit first, evaluation metrics are to be pre-registered, and each paid training run needs explicit approval.

Limitations. A molecule generator alone is not drug discovery. Binding affinity and toxicity from our scorers are model estimates, not measurements; a generated molecule is a hypothesis until it has been synthesized and measured. We expect to need partners for compound synthesis and for biochemical assays on purified targets much later in the process; we are not asking about that now.

4. Our program objectives

To build capable, locally relevant biology AI that Filipino researchers and industry can actually use; to turn Philippine resources and needs — rice-straw biomass, dengue and TB diagnostics — into concrete, testable projects; and to report results transparently, including the negative ones you have just read.

5. What is in development right now, and where it stands

LIKHA enzyme specialization is our only model with active training work today, and the findings in Section 3 are its current state: a real modeling improvement and a genuine but non-specific prompting effect, alongside honest quality and family-control gaps that we are addressing by expanding the training data. Because the model does not yet produce reliable, family-specific candidates, we are deliberately not proposing sequences for testing at this stage — we would rather design the downstream workflow properly first. HYGEIA and Molek remain at the planning stage.

6. What we are hoping for

We are looking for a wet-laboratory partner in the Philippines, and this brief is an introduction: we would like to know whether this kind of collaboration is something your group might be interested in exploring. We are not asking you to commit to anything, or to test anything, at this stage. If there is interest, we would suggest a short conversation to see whether our interests fit, and we would work out the practical details (what would be tested, how, and on what terms) together, in a way that suits your laboratory.

What we would bring. Computational candidates with full provenance (model version, sampling settings, novelty against known proteins, family evidence, and predicted-structure confidence), honest caveats about what a prediction can and cannot show, open sharing of our experimental findings and behavior analyses, and acknowledgement or co-authorship as appropriate. We would only ever ask you to test sequences that have passed our computational screening, and we expect any first collaboration to be limited to benign enzymes.